

CBSE BOARD EXAMINATION PRACTICE MODULE

Class X • Mathematics (Standard / Basic) • Chapter: Probability

Topic: Theoretical Probability & Events

Target Year: Board Exams

Max. Time: 45 Mins

SECTION A — MULTIPLE CHOICE QUESTIONS

Q.55 What is the probability that a leap year has 53 Mondays?

(a) $\frac{2}{7}$

(b) $\frac{4}{7}$

(c) $\frac{5}{7}$

(d) $\frac{6}{7}$

Q.56 If a two digit number is chosen at random, then the probability that the number chosen is a multiple of 3, is:

(a) $\frac{3}{10}$

(b) $\frac{29}{100}$

(c) $\frac{1}{3}$

(d) $\frac{7}{25}$

SECTION B — SUBJECTIVE QUESTIONS

Q.1 A bag contains 5 black, 7 red and 3 white balls. A ball is drawn from the bag at random. Find the probability that the ball drawn is:

(i) red

(ii) black or white

(iii) not black.

Q.2 A bag contains 7 black, 5 red and 3 white balls. A ball is drawn from the bag at random. Find the probability that the ball drawn is:

(i) red

(ii) black or white

(iii) not black.

Q.3 A bag contains 6 red, 5 black and 4 white balls. A ball is drawn from the bag at random. Find the probability that:

(i) white

(ii) red

(iii) not black

(iv) red or white.

Q.4 A card is drawn at random from a pack of 52 playing cards. Find the probability that the card drawn is neither an ace nor a king.

Q.5 A card is drawn at random from a pack of 52 playing cards. Find the probability that the card drawn is neither a red card nor a black king.

Q.6 Out of 400 bulbs in a box, 15 bulbs are defective. One bulb is taken out at random from the box. Find the probability that the drawn bulb is not defective.

Q.7 Find the probability of getting 53 Fridays in a leap year.

Q.8 A card is drawn at random from well shuffled pack of 52 cards. Find the probability that the card drawn is neither a red card nor queen.

Q.9 Find the probability that a number selected from number 1 to 25 is not a prime numbers when each of given numbers is equally likely to be selected.

Q.10 There are 30 cards, of same size, in a bag on which numbers 1 to 30 are written. One card is taken out of the bag at random. Find the probability that the number on the selected card is not divisible by 3.

Q.11 A bag contains 5 red, 8 white and 7 black balls. A ball is drawn from the bag at random. Find the probability that the drawn ball is:

(i) red or white

(ii) not black

(iii) neither white nor black.

Q.12 A box contains 20 cards, numbered from 1 to 20. A card is drawn from the box at random. Find the probability that the number on the card is:

(i) even

(ii) prime

(iii) multiple of 3.